

MT 303: Computational Materials Science

Course Code	MT 303
Course Name	Computational Materials Science
L-T-P-C	3-2-0-4
Prerequisites	None
Intended for	B.Tech. (MSE)
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction:** Definition and scope of Computational Materials Science; Overview of computational methods: Quantum Mechanics, Molecular Dynamics, Monte Carlo simulations; Role of computational tools in materials discovery and design; Hands-on session introducing materials modeling codes [4 Hours]
- **Quantum Mechanical Methods for Materials Simulation:** Basics of quantum mechanics applied to materials science; Density Functional Theory (DFT) and its practical applications; Quantum Monte Carlo methods and their role in materials simulations; Practical exercises using materials modeling codes for electronic structure calculations [10 Hours]
- **Classical Methods for Materials Simulation:** Molecular Dynamics simulations: principles and applications; Monte Carlo simulations in materials science: theory and practice; Introduction to classical force fields; Application of classical methods in predicting material behavior [8 Hours]
- **Electronic Structure Calculations:** Band structure calculations and their interpretation; Electronic density of states and its significance; Tight-binding and empirical methods for electronic structure predictions; Hands-on application of computational tools for electronic structure analysis [12 Hours]
- **Materials Design and Optimization:** Rational design of materials using computational methods; Optimization algorithms and their role in materials science; High-throughput screening for material discovery; Case studies on computationally designed materials [6 Hours]
- **Advanced Topics in Computational Materials Science:** Quantum transport simulations for materials applications; Computational studies of surfaces and interfaces; Machine learning applications in materials science; Future directions and challenges in the field [10 Hours]

Laboratory Modules

1. Introduction to computational software for materials science simulations.
2. Quantum mechanical simulations using DFT software.
3. Molecular Dynamics simulations for understanding material behavior.
4. Electronic structure calculations and visualization.
5. Materials design and optimization using computational tools.
6. Hands-on session on machine learning applications in materials science.

Text Books

1. Richard LeSar, **Introduction to Computational Materials Science**.
2. June Gunn Le, **Computational Materials Science: An Introduction**.
3. Koenraad George Frans Janssens, **Computational Materials Engineering: An Introduction to Microstructure Evolution**.
4. Richard M. Martin, **Electronic Structure: Basic Theory and Practical Methods**.
5. Peter Comba, **Molecular Modeling of Inorganic Compounds**.
6. David Sholl and Janice A. Steckel, **Density Functional Theory: A Practical Introduction**.
7. Keith T. Butler, Felipe Oviedo, **Pieremanuele Canepa Machine Learning in Materials Science**.

References

1. NA